Observability and Audit Evidence Framework — Validation Test Specification

OAEF — Test plan, test design, test cases, procedures and completion report for the validation of the framework once implemented and deployed, in simulation mode and in full mode on scenario NIN-001

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Approvals

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1 Introduction

1.1 Purpose

This document specifies how to validate that the Observability and Audit Evidence Framework (OAEF), once implemented and deployed, meets the requirements of OAEF-SPEC-001. Its structure follows ISO/IEC/IEEE 29119-3: test plan (chapter 2), test design (chapter 3), test environment and data (chapter 4), test procedures (chapter 5), test cases (chapter 6), traceability (chapter 7) and completion report template (chapter 8).

1.2 Validation modes

Mode	Principle	When
S — Simulation	Synthetic telemetry generated with OpenTelemetry telemetrygen, and replay of the NIN-001 OTLP example with fresh identifiers and timestamps. Validates the generic layer and the qualification layer without MOSIP.	As soon as the generic layer is deployed (WP0.1), before instrumentation of MOSIP is complete.
F — Full	Execution of NIN-001 (and selected variants) on the MOSIP test platform with Jarvis, real signals, evidence and verdict.	After WP0.2 and WP0.3; mandatory for acceptance.
I / D	Inspection of configuration and documents; demonstration of a workflow.	Any time.

1.3 References

Ref	Document
[R1]	OAEF-SPEC-001 — Observability and Audit Evidence Framework Specification.
[R2]	ISO/IEC/IEEE 29119-3:2021 — Test documentation.
[R3]	Repository mosip-observability-oracles (oracles, examples, feature, persona, prerequisites).
[R4]	OpenTelemetry Collector contrib — telemetrygen tool documentation (version pinned with the collector).

2 Test plan

2.1 Test items

- Generic layer: instrumentation conditions, collector agent and gateway, Tempo, Loki (app and audit tenants), Mimir, Grafana, object storage.
- Deployment: Terraform modules, Helm values, Argo CD applications, overlays generated from the environment manager.
- Qualification layer: Jarvis, data generator, mocks, evidence collector, oracle engine, reporter.
- Reference implementation: NIN-001 oracle, feature, persona and prerequisites.

2.2 Features to be tested

All requirements of OAEF-SPEC-001 chapter 5 (SW, TST, INF, SEC, DQ, OPS). Features not tested in this version: production monitoring service levels; performance of MOSIP itself.

2.3 Approach

1. **Phase V1 — Simulation**: VT-INF, VT-PIPE, VT-SEC (mode S), VT-QUAL. Demonstrates the pipeline, storage, security controls and verdict logic with controlled inputs, including injected faults.
2. **Phase V2 — Full NIN-001**: VT-SW, VT-E2E. Demonstrates the real instrumentation of MOSIP and the full evidence chain.
3. **Phase V3 — Inspection and demonstration**: remaining I and D test cases, sign-off.

Each test case produces its own evidence (chapter 6) stored in the evidence bucket under `validation/<campaign-id>/<test-case-id>/`.

2.4 Entry criteria

- V1: observability stack and qualification tooling deployed on the target platform by GitOps; telemetrygen available at the pinned version; credentials for read-only service accounts.
- V2: MOSIP test platform deployed; prerequisites env-nin-baseline green; MOSIP extension and audit patch deployed.

2.5 Exit criteria

- All test cases covering **M** requirements passed; no open defect of severity critical or major.
- VT-E2E-01 passed twice consecutively on TEST with identical verdicts.
- Every requirement of OAEF-SPEC-001 covered by at least one passed test case (chapter 7).

2.6 Suspension and resumption

Suspend when the pipeline-health objectives are breached (results would be INCONCLUSIVE) or when the platform drifts from the environment manager. Resume after correction and a passing VT-INF-01 canary.

2.7 Roles

Role	Responsibility
Test lead	Plans, runs campaigns, signs the completion report.
Platform engineer	Deploys, injects infrastructure faults, restores.
Observability owner	Reviews signal and oracle results.
Security officer	Witnesses VT-SEC tests and approves.

Role	Responsibility
Functional owner	Reviews NIN-001 functional results and golden runs.

2.8 Risks

Risk	Mitigation
Configuration keys differ in the pinned versions	Validate configuration baselines in DEV first; record versions in each test record.
MOSIP extension not ready for V2	Run V1 fully; mark dependent VT-SW cases blocked, not failed.
Shared platform noise	Use ephemeral namespaces (VT-INF-09) or relative assertions.

3 Test design

Feature set	Covers	Test cases
FS-1 Infrastructure	Stack per site, WORM, tenancy, capacity, time, network, drift, GitOps, DR	VT-INF-01 to VT-INF-10
FS-2 Pipeline	Ingestion, sampling, redaction, routing, derived metrics, health, business events	VT-PIPE-01 to VT-PIPE-07
FS-3 Instrumentation	Agent, keys, propagation, links, audit correlation, metrics, edges, business events, profiling	VT-SW-01 to VT-SW-11
FS-4 Security	PII, lookup, access, transport, baggage, integrity, compliance, supply chain, publication	VT-SEC-01 to VT-SEC-09
FS-5 Data quality	Registry, contract, cardinality	VT-DQ-01 to VT-DQ-03
FS-6 Qualification layer	Root span, evidence, oracle, release awareness, data, mocks, reports, verdicts	VT-QUAL-01 to VT-QUAL-08
FS-7 End to end	NIN-001, audit gap, DR site	VT-E2E-01 to VT-E2E-03
FS-8 Operations	Dashboards, alerts, governance	VT-OPS-01, VT-OPS-02

4 Test environment and data

4.1 Environment

Element	Requirement
Platform	TEST (and DC2 for VT-E2E-03, VT-INF-10)
Tools	telemetrygen (pinned), curl, Python 3, a TraceQL/LogQL/PromQL client or Grafana, S3 client, Jarvis
Accounts	Read-only service accounts per tenant; auditor test user; non-privileged test user
Variables	`AGENT` collector agent host, `TEMPO`, `LOKI`, `MIMIR`, `GRAFANA` base URLs

4.2 Test data

- **D1 — NIN-001 OTLP example**: `oracles/examples/NIN-001.otlp-example.json` (7 spans, 1 log record, 1 histogram data point with exemplar).
- **D2 — Synthetic load**: telemetrygen profiles per platform (rates [X], [Y], [Z]).
- **D3 — Fault data sets**: variants of D1 for verdict classification: missing span; error span with exception event; Kubernetes restart event in the window; business rejection code; wrong outcome without error signal; pipeline breach.
- **D4 — PII seeds**: fake 11-digit NIN, fake email and phone, never real values.
- **D5 — Persona adult-nohit** for full mode.
- **D6 — NIN-001 business events**: the six catalogue events of OAEF-SW-12 as OTLP log records with event.name, attached to the spans of D1 (same trace_id and span_id).

5 Test procedures

5.1 TP-1 Replay of the NIN-001 OTLP example (simulation)

The example must be replayed with new trace and span identifiers and current timestamps, so that each replay is unique and inside the retention window. Python handles the 64-bit nanosecond integers exactly.

```
#!/usr/bin/env python3
# replay_nin001.py  —  usage: python3 replay_nin001.py NIN-001.otlp-example.json http://AGENT:4318
import json, os, sys, time, urllib.request
src, endpoint = sys.argv[1], sys.argv[2].rstrip('/')
doc = json.load(open(src, encoding='utf-8'))
ids = {}
def new_id(old, n):
    if not old: return old
    return ids.setdefault(old, os.urandom(n).hex())
t_min = min(int(s['startTimeUnixNano']) for rs in doc['traces']['resourceSpans']
            for ss in rs['scopeSpans'] for s in ss['spans'])
shift = time.time_ns() - t_min - 60_000_000_000          # start one minute ago
def fix(o):
    if isinstance(o, dict):
        for k, v in list(o.items()):
            if k.endswith('UnixNano') and isinstance(v, str): o[k] = str(int(v) + shift)
            elif k == 'traceId': o[k] = new_id(v, 16)
            elif k in ('spanId', 'parentSpanId'): o[k] = new_id(v, 8)
            else: fix(v)
    elif isinstance(o, list):
        for x in o: fix(x)
for block in ('traces', 'logs', 'metrics'): fix(doc[block])
for block, path in (('traces', '/v1/traces'), ('logs', '/v1/logs'), ('metrics', '/v1/metrics')):
    req = urllib.request.Request(endpoint + path, data=json.dumps(doc[block]).encode(),
                                 headers={'Content-Type': 'application/json'})
    print(block, urllib.request.urlopen(req).status)
root = ids['4bf92f3577b34da6a3ce929d0e0e4736']
print('replayed trace id:', root)
```

Verification queries (replace `<trace_id>` with the printed id):

```
curl -s "$TEMPO/api/v2/traces/<trace_id>"                       # 7 spans, ids returned base64: normalise to
hex
curl -s -G "$LOKI/loki/api/v1/query_range" -H "X-Scope-OrgID: app" \
     --data-urlencode 'query={service_name="regproc-pkr"} | trace_id="<trace_id>"'
curl -s -G "$MIMIR/prometheus/api/v1/query_exemplars" \
     --data-urlencode 'query=http_server_request_duration_seconds_bucket{job="mosip/regproc-pkr"}' \
     --data-urlencode "start=$(date -d '-10 min' +%s)" --data-urlencode "end=$(date +%s)"
```

5.2 TP-2 Synthetic telemetry with telemetrygen

```
# traces: 100 test-tagged traces with 5 child spans, wrong site on purpose (stamping test)
telemetrygen traces --otlp-endpoint $AGENT:4317 --otlp-insecure --traces 100 --child-spans 5 \
  --service regproc-pkr \
  --otlp-attributes 'deployment.environment.name="wrong-env"' --otlp-attributes 'cloud.region="dcx"' \
  --telemetry-attributes 'test.case.name="VT-INF-03"' --telemetry-attributes
'mosip.rid="10001100010000120261018102233"'

# logs: audit stream records for routing test
telemetrygen logs --otlp-endpoint $AGENT:4317 --otlp-insecure --logs 20 \
  --body 'AUDIT UIN_GENERATOR ref_id=10001100010000120261018102233' \
  --telemetry-attributes 'stream="mosip-audit"'

# load: 1.5x nominal for 30 minutes
telemetrygen traces --otlp-endpoint $AGENT:4317 --otlp-insecure --rate [1.5*X] --duration 30m --service load-
test
```

Flag names follow telemetrygen of the pinned collector release and must be checked against it before the campaign.

5.3 TP-3 Fault injection for verdict classification

Data set (from D1)	Modification	Expected verdict
D3-a complete	None	PASS
D3-b missing span	Remove the idgenerator server span	PASS with evidence gap (functional stub OK) or SILENT FAILURE (functional stub KO)
D3-c error span	Status code 2 and an exception event on the identity span; functional stub KO	PRODUCT BUG
D3-d infra event	Pod restart (OOMKilled) recorded in the window; functional stub KO	INFRA
D3-e config rejection	Business rejection code for inactive centre; functional stub KO	TEST DATA / CONFIG
D3-f pipeline breach	Gateway scaled to 0 during the window	INCONCLUSIVE

5.4 TP-4 Full NIN-001 run

1. Resolve the target environment (for example `test-int@DC1`) and run the prerequisite checks.
2. Execute `features/nin_001_new_adult_no_hit.feature` with Jarvis.
3. Let Jarvis settle, collect, seal and verify; archive the report and the evidence bundle.
4. Verify the bundle signature and hashes independently.

5.5 TP-5 Evidence bundle verification

```
sha256sum -c manifest.sha256        # or the verification command of the evidence collector
# verify signature.jws against the published verification key, then recompute the verdict offline
```

6 Test cases

Each test case lists the requirements it covers, the mode, preconditions, steps, expected result and the evidence to retain. Result fields are completed during execution.

6.0 Test case summary

ID	Title	Mode	Requirements
VT-INF-01	Canary signal end to end on every site and platform	S	OAEF-INF-01, OAEF-OPS-01
VT-INF-02	Object lock on audit and evidence buckets	S	OAEF-INF-02, OAEF-SEC-06
VT-INF-03	Environment and site stamping overrides application values	S	OAEF-INF-03, OAEF-INF-07
VT-INF-04	Ingest capacity at 1.5x nominal load	S	OAEF-INF-04, OAEF-DQ-04
VT-INF-05	Clock synchronisation	I	OAEF-INF-05
VT-INF-06	Network exposure of OTLP and runner egress	S	OAEF-INF-06, OAEF-INF-10
VT-INF-07	Environment manager drift detection	S	OAEF-INF-07
VT-INF-08	GitOps reconciliation	S	OAEF-INF-08
VT-INF-09	Ephemeral namespace with exact metric deltas	F	OAEF-INF-09
VT-INF-10	Backup, restore and DC2 evidence availability	S	OAEF-INF-11
VT-PIPE-01	OTLP replay of the NIN-001 example through the full pipeline	S	OAEF-TST-02, OAEF-DQ-04, OAEF-SW-11
VT-PIPE-02	Tail sampling keeps test and error traces	S	OAEF-DQ-04
VT-PIPE-03	PII redaction at the gateway	S	OAEF-SEC-01
VT-PIPE-04	Audit routing and multiset equality	S	OAEF-SW-05, OAEF-SEC-03
VT-PIPE-05	Span metrics and exemplars	S	OAEF-SW-07, OAEF-DQ-03
VT-PIPE-06	Pipeline failure leads to INCONCLUSIVE	S	OAEF-OPS-01, OAEF-TST-03
VT-PIPE-07	Business events through the pipeline	S	OAEF-SW-12, OAEF-DQ-01
VT-SW-01	Agent present and pinned on every component	F	OAEF-SW-01, OAEF-SW-11
VT-SW-02	Business keys on spans and logs, no clear identifiers	F	OAEF-SW-02, OAEF-SW-06
VT-SW-03	Propagation across Kafka and JMS stages	F	OAEF-SW-03
VT-SW-04	Span links for credential batch and WebSub	F	OAEF-SW-04
VT-SW-05	Audit trace correlation	F	OAEF-SW-05
VT-SW-06	Business metrics present	F	OAEF-SW-07
VT-SW-07	Offline station trace continuity	F	OAEF-SW-08, OAEF-INF-05
VT-SW-08	Front-end root span	F	OAEF-SW-09
VT-SW-09	Partner edge: remote parent and partner id	S	OAEF-SW-10, OAEF-SEC-05
VT-SW-10	Business event sequence and RID-to-NIN duration	F	OAEF-SW-12, OAEF-SW-04, OAEF-SW-05
VT-SW-11	Profile attached to a span over budget	S	OAEF-SW-13
VT-SEC-01	PII scan negative test	S	OAEF-SEC-01
VT-SEC-02	NIN lookup helper restricted and audited	D	OAEF-SEC-02
VT-SEC-03	Access control matrix	S	OAEF-SEC-03
VT-SEC-04	Transport security and secrets	S	OAEF-SEC-04
VT-SEC-05	Baggage hygiene	S	OAEF-SEC-05

ID	Title	Mode	Requirements
VT-SEC-06	Evidence tamper detection	S	OAEF-SEC-06, OAEF-TST-02
VT-SEC-07	Data protection inspection	I	OAEF-SEC-07
VT-SEC-08	Supply-chain admission	S	OAEF-SEC-08
VT-SEC-09	Public repository review	I	OAEF-SEC-09
VT-DQ-01	Attribute registry lint	S	OAEF-DQ-01
VT-DQ-02	Signal contract in Definition of Done	D	OAEF-DQ-02
VT-DQ-03	Cardinality budget	S	OAEF-DQ-03
VT-QUAL-01	ROOT span and baggage	S	OAEF-TST-01
VT-QUAL-02	Evidence bundle completeness and replay	S	OAEF-TST-02
VT-QUAL-03	Oracle schema and golden-run workflow	S	OAEF-TST-03
VT-QUAL-04	Release-aware oracle	S	OAEF-TST-04
VT-QUAL-05	Synthetic data generator	S	OAEF-TST-05
VT-QUAL-06	Mock modes	S	OAEF-TST-06
VT-QUAL-07	Report links	D	OAEF-TST-07
VT-QUAL-08	Verdict classification with injected faults	S	OAEF-TST-03, OAEF-OPS-01
VT-E2E-01	NIN-001 full run	F	OAEF-SW-01, OAEF-SW-02, OAEF-SW-03, OAEF-SW-05, OAEF-SW-06, OAEF-SW-12, OAEF-TST-01, OAEF-TST-02, OAEF-TST-03, OAEF-INF-03, OAEF-SEC-01
VT-E2E-02	NIN-091 audit gap detected	F	OAEF-SW-05, OAEF-DQ-02
VT-E2E-03	NIN-080 run on DC2	F	OAEF-INF-03, OAEF-INF-11
VT-OPS-01	Dashboards and alerts provisioned	I	OAEF-OPS-02
VT-OPS-02	Governance and support artefacts	I	OAEF-OPS-03, OAEF-OPS-04, OAEF-OPS-05

6.1 Infrastructure (FS-1)

VT-INF-01 Canary signal end to end on every site and platform

Requirements	OAEF-INF-01, OAEF-OPS-01
Mode	S — Simulation
Preconditions	Stack deployed on DC1 and DC2; canary job configured.
Steps	1. Emit one canary span, log and metric per node with telemetrygen (resource attributes of the node). 2. Query Tempo, Loki and Mimir for the canary ids. 3. Measure the delay between emission and availability.
Expected result	All canary signals retrievable on each site within [60 s]; site and environment attributes present.
Evidence to retain	Canary ids, query results, timings.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-02 Object lock on audit and evidence buckets

Requirements	OAEF-INF-02, OAEF-SEC-06
Mode	S — Simulation
Preconditions	Buckets created by Terraform with object lock.
Steps	1. Upload a test object to the evidence bucket with retention. 2. Attempt delete and overwrite with an administrator role. 3. Check the replica on DC2.
Expected result	Delete and overwrite refused; replica present on DC2.
Evidence to retain	API responses, bucket policy export.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-03 Environment and site stamping overrides application values

Requirements	OAEF-INF-03, OAEF-INF-07
Mode	S — Simulation
Preconditions	Collector agent with site stamp from the environment manager.
Steps	1. Send spans with telemetrygen declaring a wrong deployment.environment.name and cloud.region. 2. Query the stored spans.
Expected result	Stored resource attributes equal the environment-manager values, not the sent ones.
Evidence to retain	Sent payload, stored span.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-04 Ingest capacity at 1.5x nominal load

Requirements	OAEF-INF-04, OAEF-DQ-04
Mode	S — Simulation
Preconditions	Agreed nominal rates [X], [Y], [Z].
Steps	1. Generate traces, logs and metrics with telemetrygen at 1.5x nominal for [30 min]. 2. Watch collector refused / failed counters and ingestion lag.
Expected result	Refused and failed counters stay at 0; lag within objective; test-tagged traces all kept.
Evidence to retain	Collector self-metrics, load profile.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-05 Clock synchronisation

Requirements	OAEF-INF-05
Mode	I — Inspection
Preconditions	NTP / chrony configured.
Steps	1. Read the clock offset metric of every node, station and simulator.
Expected result	Offset at most [50 ms] everywhere.
Evidence to retain	Offset metrics export.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-06 Network exposure of OTLP and runner egress

Requirements	OAEF-INF-06, OAEF-INF-10
Mode	S — Simulation
Preconditions	Network policies applied.
Steps	1. Attempt OTLP connection from outside the cluster. 2. From a CI runner, attempt a connection to a non-allowed destination.
Expected result	Both connections refused; allowed destinations reachable.
Evidence to retain	Connection logs.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-07 Environment manager drift detection

Requirements	OAEF-INF-07
Mode	S — Simulation
Preconditions	Sync job and drift check deployed.
Steps	1. Change the site value of one collector manually. 2. Run the drift check.
Expected result	Drift reported with the exact attribute and cluster; campaign blocked (ENV_NOT_READY).
Evidence to retain	Drift report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-08 GitOps reconciliation

Requirements	OAEF-INF-08
Mode	S — Simulation
Preconditions	Argo CD managing the observability namespace.
Steps	1. Edit a collector config map directly in the cluster. 2. Wait for the next sync.
Expected result	Manual change reverted; drift alarm raised.
Evidence to retain	Sync history.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-09 Ephemeral namespace with exact metric deltas

Requirements	OAEF-INF-09
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Ephemeral namespace template available.
Steps	1. Create an ephemeral MOSIP slice. 2. Run NIN-001 with metric assertions set to exact equality.
Expected result	NIN-001 PASS with exact deltas.
Evidence to retain	Evidence bundle.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-INF-10 Backup, restore and DC2 evidence availability

Requirements	OAEF-INF-11
Mode	S — Simulation
Preconditions	Backup jobs configured.
Steps	1. Restore observability stores from backup on a test cluster. 2. Read an evidence bundle from DC2 with DC1 endpoints disabled.
Expected result	Restore succeeds; bundle verifies from DC2.
Evidence to retain	Restore log, verification output.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.2 Pipeline (FS-2)

VT-PIPE-01 OTLP replay of the NIN-001 example through the full pipeline

Requirements	OAEF-TST-02, OAEF-DQ-04, OAEF-SW-11
Mode	S — Simulation
Preconditions	Example file oracles/examples/NIN-001.otlp-example.json; timestamps shifted to now.
Steps	1. POST the traces, logs and metrics blocks to the agent OTLP/HTTP endpoints /v1/traces, /v1/logs, /v1/metrics. 2. Retrieve the trace by id from Tempo, the log record from Loki and the metric and exemplar from Mimir. 3. Normalise Tempo ids from base64 to hex and compare with the sent payload.
Expected result	All 7 spans, the log record and the histogram data point retrieved; parent links intact; exemplar points to span 53995c3f42cd8ad8.
Evidence to retain	Replay payload, query results, comparison report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-PIPE-02 Tail sampling keeps test and error traces

Requirements	OAEF-DQ-04
Mode	S — Simulation
Preconditions	PERF sampling policy active.
Steps	1. Send 1 000 untagged traces, 100 traces with test.case.name, 50 traces with ERROR status. 2. Count stored traces per category.
Expected result	All test-tagged and error traces stored; untagged traces sampled at about [10 %].
Evidence to retain	Counts per category.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-PIPE-03 PII redaction at the gateway

Requirements	OAEF-SEC-01
Mode	S — Simulation
Preconditions	Redaction processor configured.
Steps	1. Send a span and a log record containing a fake 11-digit NIN, an email and a phone number in attributes and body. 2. Query the stored records.
Expected result	Values masked or removed in storage; redaction counter incremented.
Evidence to retain	Sent and stored records.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-PIPE-04 Audit routing and multiset equality

Requirements	OAEF-SW-05, OAEF-SEC-03
Mode	S — Simulation
Preconditions	Routing to the audit tenant configured.
Steps	1. Send 20 audit log records (stream mosip-audit) and 20 application records. 2. Query both tenants with and without the auditor role.
Expected result	Audit records only in the audit tenant; readable by auditors only; counts equal to sent.
Evidence to retain	Query results per role.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-PIPE-05 Span metrics and exemplars

Requirements	OAEF-SW-07, OAEF-DQ-03
Mode	S — Simulation
Preconditions	Spanmetrics connector and exemplars enabled.
Steps	1. Replay NIN-001 spans. 2. Query traces_span_metrics_calls_total and query_exemplars for the packet-receiver route.
Expected result	Counter incremented with environment and site labels; exemplar carries the replayed trace id; no identifier used as a label.
Evidence to retain	PromQL results.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-PIPE-06 Pipeline failure leads to INCONCLUSIVE

Requirements	OAEF-OPS-01, OAEF-TST-03
Mode	S — Simulation
Preconditions	Pipeline health objectives configured.
Steps	1. Scale the gateway to 0 replicas. 2. Run a simulated scenario and the evidence collector. 3. Restore the gateway.
Expected result	Verdict INCONCLUSIVE with pipeline-health reason; never PASS.
Evidence to retain	verdict.json, health metrics.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-PIPE-07 Business events through the pipeline

Requirements	OAEF-SW-12, OAEF-DQ-01
Mode	S — Simulation
Preconditions	Replay data D6 (catalogue events of NIN-001 inside the D1 spans) available.
Steps	1. POST the D6 log records to /v1/logs, then one event with an event.name absent from the catalogue and one carrying a fake NIN. 2. Query Loki for event.name and join each event to its span by trace_id and span_id.
Expected result	All catalogue events stored with event.name, trace_id and span_id and joined to their spans; unknown event name flagged by the contract check; fake NIN redacted.
Evidence to retain	Replay payload, query results, contract report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.3 Instrumentation (FS-3)

VT-SW-01 Agent present and pinned on every component

Requirements	OAEF-SW-01, OAEF-SW-11
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Platform deployed.
Steps	1. List pods with injected agent and versions. 2. Read the evidence manifest of a NIN-001 run.
Expected result	Every inventory component instrumented with the pinned version; manifest lists versions and digests.
Evidence to retain	Inventory export, manifest.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-02 Business keys on spans and logs, no clear identifiers

Requirements	OAEF-SW-02, OAEF-SW-06
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Extension deployed.
Steps	1. Run NIN-001. 2. Check mosip.rid and mosip.uin_hash on all expected spans and trace_id / span_id on log records. 3. Run the PII scan.
Expected result	Keys present on 100 % of expected spans; logs correlated; PII scan 0.
Evidence to retain	Oracle check results.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-03 Propagation across Kafka and JMS stages

Requirements	OAEF-SW-03
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	NIN-001 executable.
Steps	1. Run NIN-001. 2. Check that every stage consumer span descends from the packet-receiver span.
Expected result	Single trace T1 across all stages, ABIS included or linked.
Evidence to retain	Trace export.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-04 Span links for credential batch and WebSub

Requirements	OAEF-SW-04
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Extension and patch deployed.
Steps	1. Run NIN-001. 2. Resolve links of T2 and T3.
Expected result	Both links resolve to spans of T1.
Evidence to retain	Trace exports.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-05 Audit trace correlation

Requirements	OAEF-SW-05
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Audit patch deployed.
Steps	1. Run NIN-001. 2. Join audit rows to spans by trace_id and span_id.
Expected result	Every audit row joined; database and audit-tenant counts equal.
Evidence to retain	Join report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-06 Business metrics present

Requirements	OAEF-SW-07
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Metrics implemented.
Steps	1. Run NIN-001. 2. Query each agreed business metric.
Expected result	All metrics exist with agreed names, units and labels and move as expected.
Evidence to retain	PromQL results.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-07 Offline station trace continuity

Requirements	OAEF-SW-08, OAEF-INF-05
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Instrumented registration client.
Steps	1. Enrol a resident with the station offline for the agreed window. 2. Reconnect and let it sync.
Expected result	Station spans and central spans share one trace id; no packet or telemetry lost.
Evidence to retain	Trace export, station buffer log.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-08 Front-end root span

Requirements	OAEF-SW-09
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Instrumented resident portal.
Steps	1. Run the resident update journey from a browser.
Expected result	Browser span is root of the backend trace; no personal data in browser attributes.
Evidence to retain	Trace export.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-09 Partner edge: remote parent and partner id

Requirements	OAEF-SW-10, OAEF-SEC-05
Mode	S — Simulation
Preconditions	Relying-party simulator.
Steps	1. Call the authentication API from an authorised simulator with a traceparent, then from an unauthorised one. 2. Capture egress towards the simulator.
Expected result	Authorised call: remote parent kept and mosip.partner_id set; unauthorised: new trace; no baggage on egress.
Evidence to retain	Traces, egress capture.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-10 Business event sequence and RID-to-NIN duration

Requirements	OAEF-SW-12, OAEF-SW-04, OAEF-SW-05
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Event catalogue implemented in the MOSIP extension; span links deployed.
Steps	1. Run NIN-001. 2. Check the oracle section events: catalogue order, one producing span per event, business keys, no personal data. 3. Compute the duration from mosip.registration.packet.received to mosip.notification.sent for the RID.
Expected result	All events present in order and joined to spans of T1, T2 and T3; duration within NF-04; audit rows match the audit events.
Evidence to retain	Event export, join report, duration.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SW-11 Profile attached to a span over budget

Requirements	OAEF-SW-13
Mode	S — Simulation
Preconditions	PERF platform with profiler; test service with a configurable CPU-bound endpoint.
Steps	1. Call the endpoint so that its server span exceeds a budget of [500 ms]. 2. Run the evidence collector on the trace.
Expected result	The bundle holds the profile of the service for the span window; the top frames include the CPU-bound method; the verdict is unchanged by the profile.
Evidence to retain	Profile file, bundle manifest.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.4 Security (FS-4)

VT-SEC-01 PII scan negative test

Requirements	OAEF-SEC-01
Mode	S — Simulation
Preconditions	PII scan job.
Steps	1. Bypass the redaction on a test pipeline and send a seeded fake NIN. 2. Run the PII scan.
Expected result	Seeded value detected and reported.
Evidence to retain	Scan report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-02 NIN lookup helper restricted and audited

Requirements	OAEF-SEC-02
Mode	D — Demonstration
Preconditions	Lookup helper deployed.
Steps	1. Search one NIN as an investigator. 2. Attempt the same as a user without the role. 3. Inspect the lookup audit.
Expected result	Authorised search returns journeys; unauthorised refused; both attempts audited; salt not exposed.
Evidence to retain	Audit records, screenshots.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-03 Access control matrix

Requirements	OAEF-SEC-03
Mode	S — Simulation
Preconditions	SSO and roles configured.
Steps	1. For each role, query each tenant and the evidence bucket.
Expected result	Results match the access matrix; denials logged.
Evidence to retain	Matrix results.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-04 Transport security and secrets

Requirements	OAEF-SEC-04
Mode	S — Simulation
Preconditions	TLS enabled.
Steps	1. Attempt a plain-text OTLP and API connection. 2. Run a secret scan on all repositories.
Expected result	Plain text refused; no secret found.
Evidence to retain	Connection log, scan report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-05 Baggage hygiene

Requirements	OAEF-SEC-05
Mode	S — Simulation
Preconditions	Egress gateway rule.
Steps	1. Send a request with baggage that triggers a call to a partner simulator. 2. Send an oversized baggage header.
Expected result	No baggage on egress; oversized baggage rejected or truncated.
Evidence to retain	Egress capture.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-06 Evidence tamper detection

Requirements	OAEF-SEC-06, OAEF-TST-02
Mode	S — Simulation
Preconditions	Sealed bundle from a simulated run.
Steps	1. Modify one byte of logs.ndjson in a copy of the bundle. 2. Run bundle verification.
Expected result	Verification fails naming the altered file.
Evidence to retain	Verification output.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-07 Data protection inspection

Requirements	OAEF-SEC-07
Mode	I — Inspection
Preconditions	DPIA and retention configuration.
Steps	1. Inspect the DPIA, data sources and retention settings.
Expected result	Synthetic data only; DPIA approved; retention configured as agreed.
Evidence to retain	Inspection record.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-08 Supply-chain admission

Requirements	OAEF-SEC-08
Mode	S — Simulation
Preconditions	Admission policy enabled.
Steps	1. Deploy an unsigned collector image.
Expected result	Deployment rejected.
Evidence to retain	Admission log.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-SEC-09 Public repository review

Requirements	OAEF-SEC-09
Mode	I — Inspection
Preconditions	Repository public.
Steps	1. Run a scan for URLs, hostnames, keys and identifiers; manual review.
Expected result	No sensitive content; review recorded.
Evidence to retain	Scan and review record.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.5 Data quality (FS-5)

VT-DQ-01 Attribute registry lint

Requirements	OAEF-DQ-01
Mode	S — Simulation
Preconditions	Lint job in CI.
Steps	1. Submit a change emitting an unknown custom attribute.
Expected result	CI fails naming the attribute.
Evidence to retain	CI log.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-DQ-02 Signal contract in Definition of Done

Requirements	OAEF-DQ-02
Mode	D — Demonstration
Preconditions	Work-package pipelines.
Steps	1. Show the contract job in each pipeline; break a contract on purpose.
Expected result	Pipeline blocked.
Evidence to retain	Pipeline runs.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-DQ-03 Cardinality budget

Requirements	OAEF-DQ-03
Mode	S — Simulation
Preconditions	Budget agreed.
Steps	1. Count active series per service after a NIN-001 run and a load test.
Expected result	Under budget; no identifier label.
Evidence to retain	Series counts.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.6 Qualification layer (FS-6)

VT-QUAL-01 ROOT span and baggage

Requirements	OAEF-TST-01
Mode	S — Simulation
Preconditions	Jarvis configured.
Steps	1. Run a simulated scenario. 2. Inspect ROOT span and child spans.
Expected result	One ROOT span with required attributes; baggage values on all child spans.
Evidence to retain	Trace export.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-02 Evidence bundle completeness and replay

Requirements	OAEF-TST-02
Mode	S — Simulation
Preconditions	Replay data loaded (VT-PIPE-01).
Steps	1. Run the evidence collector for the replayed trace. 2. Recompute the verdict from the bundle only, offline.
Expected result	All files present with hashes and signature; recomputed verdict identical.
Evidence to retain	Bundle, recomputed verdict.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-03 Oracle schema and golden-run workflow

Requirements	OAEF-TST-03
Mode	S — Simulation
Preconditions	CI with schemas.
Steps	1. Submit an invalid oracle. 2. Submit a golden run without approval.
Expected result	Both rejected.
Evidence to retain	CI logs.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-04 Release-aware oracle

Requirements	OAEF-TST-04
Mode	S — Simulation
Preconditions	Oracle pinned to a release.
Steps	1. Run against a platform declaring another MOSIP release or agent version.
Expected result	Stops with ENV_NOT_READY before execution.
Evidence to retain	Run log.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-05 Synthetic data generator

Requirements	OAEF-TST-05
Mode	S — Simulation
Preconditions	Live ID schema.
Steps	1. Generate 1 000 residents with persona adult-nohit. 2. Validate against the ID schema; check uniqueness.
Expected result	100 % valid; no duplicates.
Evidence to retain	Validation report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-06 Mock modes

Requirements	OAEF-TST-06
Mode	S — Simulation
Preconditions	Mocks deployed.
Steps	1. Switch the ABIS mock through NO_HIT, HIT, TIMEOUT, FAILURE and call it. 2. Read SMTP mock inbox API.
Expected result	Each mode behaves as specified and appears in traces.
Evidence to retain	Traces, API responses.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-07 Report links

Requirements	OAEF-TST-07
Mode	D — Demonstration
Preconditions	Failed simulated run.
Steps	1. Open the report; follow each evidence link and the Grafana link.
Expected result	Links resolve to the right file, trace and time window.
Evidence to retain	Screenshots.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-QUAL-08 Verdict classification with injected faults

Requirements	OAEF-TST-03, OAEF-OPS-01
Mode	S — Simulation
Preconditions	Simulation data sets per class.
Steps	1. Replay data sets: complete; missing span; error span with exception; restart event in window; business rejection code; no error signal with wrong outcome; pipeline breach.
Expected result	Verdicts: PASS; PASS evidence gap; PRODUCT BUG; INFRA; TEST DATA / CONFIG; SILENT FAILURE; INCONCLUSIVE.
Evidence to retain	verdict.json per data set.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.7 End to end (FS-7)

VT-E2E-01 NIN-001 full run

Requirements	OAEF-SW-01, OAEF-SW-02, OAEF-SW-03, OAEF-SW-05, OAEF-SW-06, OAEF-SW-12, OAEF-TST-01, OAEF-TST-02, OAEF-TST-03, OAEF-INF-03, OAEF-SEC-01
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	All prerequisites of env-nin-baseline green.
Steps	1. Execute features/nin_001_new_adult_no_hit.feature on TEST. 2. Let Jarvis collect, seal and verify.
Expected result	Verdict PASS; signal contract 100 %; upload to PROCESSED within limit; PII scan 0; audit multiset equal.
Evidence to retain	Sealed evidence bundle, report.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-E2E-02 NIN-091 audit gap detected

Requirements	OAEF-SW-05, OAEF-DQ-02
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	Audit row suppression for one stage on a test build.
Steps	1. Run NIN-001 with the suppressed audit event.
Expected result	Verdict PASS with evidence gap naming the missing audit event.
Evidence to retain	verdict.json.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-E2E-03 NIN-080 run on DC2

Requirements	OAEF-INF-03, OAEF-INF-11
Mode	F — Full (NIN-001 on MOSIP)
Preconditions	DC2 primary for the test.
Steps	1. Run NIN-001 against DC2.
Expected result	PASS; every MOSIP signal carries site DC2 and role primary; none from DC1.
Evidence to retain	Evidence bundle.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

6.8 Operations (FS-8)

VT-OPS-01 Dashboards and alerts provisioned

Requirements	OAEF-OPS-02
Mode	I — Inspection
Preconditions	GitOps repository.
Steps	1. Compare dashboards and alert rules in Grafana with Git.
Expected result	Identical; alerts routed.
Evidence to retain	Inspection record.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

VT-OPS-02 Governance and support artefacts

Requirements	OAEF-OPS-03, OAEF-OPS-04, OAEF-OPS-05
Mode	I — Inspection
Preconditions	Programme documentation.
Steps	1. Inspect roles, upgrade checklist, runbooks, support rota, onboarding records.
Expected result	All present and current.
Evidence to retain	Inspection record.
Result	[ ] Pass [ ] Fail [ ] Blocked Executed by: [ ] Date: [ ] Defect: [ ]

7 Traceability

Requirement of OAEF-SPEC-001 to validation test cases.

Requirement	Pri	Ver.	Validation test cases (OAEF-VTS-001)
OAEF-SW-01	M	T,I	VT-SW-01, VT-E2E-01
OAEF-SW-02	M	T	VT-SW-02, VT-E2E-01
OAEF-SW-03	M	T	VT-SW-03, VT-E2E-01
OAEF-SW-04	S	T	VT-SW-04, VT-SW-10
OAEF-SW-05	M	T	VT-PIPE-04, VT-SW-05, VT-SW-10, VT-E2E-01, VT-E2E-02
OAEF-SW-06	M	T	VT-SW-02, VT-E2E-01
OAEF-SW-07	S	T,I	VT-PIPE-05, VT-SW-06
OAEF-SW-08	M	T	VT-SW-07
OAEF-SW-09	S	T	VT-SW-08
OAEF-SW-10	S	T	VT-SW-09
OAEF-SW-11	M	I	VT-PIPE-01, VT-SW-01
OAEF-SW-12	M	T	VT-PIPE-07, VT-SW-10, VT-E2E-01
OAEF-SW-13	S	T	VT-SW-11
OAEF-TST-01	M	T	VT-QUAL-01, VT-E2E-01
OAEF-TST-02	M	T	VT-PIPE-01, VT-SEC-06, VT-QUAL-02, VT-E2E-01
OAEF-TST-03	M	T	VT-PIPE-06, VT-QUAL-03, VT-QUAL-08, VT-E2E-01
OAEF-TST-04	M	T	VT-QUAL-04
OAEF-TST-05	M	T	VT-QUAL-05
OAEF-TST-06	M	T	VT-QUAL-06
OAEF-TST-07	S	D	VT-QUAL-07
OAEF-INF-01	M	T	VT-INF-01
OAEF-INF-02	M	T	VT-INF-02
OAEF-INF-03	M	T	VT-INF-03, VT-E2E-01, VT-E2E-03
OAEF-INF-04	M	T,A	VT-INF-04
OAEF-INF-05	M	T,I	VT-INF-05, VT-SW-07
OAEF-INF-06	M	T	VT-INF-06
OAEF-INF-07	M	T	VT-INF-03, VT-INF-07
OAEF-INF-08	M	T,I	VT-INF-08
OAEF-INF-09	S	T	VT-INF-09
OAEF-INF-10	M	T	VT-INF-06
OAEF-INF-11	S	T	VT-INF-10, VT-E2E-03
OAEF-SEC-01	M	T	VT-PIPE-03, VT-SEC-01, VT-E2E-01
OAEF-SEC-02	M	D,I	VT-SEC-02
OAEF-SEC-03	M	T	VT-PIPE-04, VT-SEC-03
OAEF-SEC-04	M	T	VT-SEC-04
OAEF-SEC-05	M	T	VT-SW-09, VT-SEC-05
OAEF-SEC-06	M	T	VT-INF-02, VT-SEC-06
OAEF-SEC-07	M	I	VT-SEC-07

Requirement	Pri	Ver.	Validation test cases (OAEF-VTS-001)
OAEF-SEC-08	S	T	VT-SEC-08
OAEF-SEC-09	M	I	VT-SEC-09
OAEF-DQ-01	M	T,I	VT-PIPE-07, VT-DQ-01
OAEF-DQ-02	M	D	VT-DQ-02, VT-E2E-02
OAEF-DQ-03	M	A,T	VT-PIPE-05, VT-DQ-03
OAEF-DQ-04	M	T	VT-INF-04, VT-PIPE-01, VT-PIPE-02
OAEF-OPS-01	M	T	VT-INF-01, VT-PIPE-06, VT-QUAL-08
OAEF-OPS-02	S	I	VT-OPS-01
OAEF-OPS-03	M	I	VT-OPS-02
OAEF-OPS-04	S	I	VT-OPS-02
OAEF-OPS-05	S	D	VT-OPS-02

8 Test completion report (template)

Item	Content
Campaign identifier	[ ]
Platform, site, versions	[environment, site, MOSIP release, agent, collector, backend versions]
Period	[from — to]
Test cases executed / passed / failed / blocked	[ ] / [ ] / [ ] / [ ]
Requirements covered by a passed test	[ ] of [ ] (must = M: all)
Open defects by severity	[critical / major / minor]
Deviations from this specification	[ ]
Evidence location	validation/<campaign-id>/
Recommendation	[accept / accept with reservations / reject]

Sign-off

Role	Name	Date	Signature
Test lead			
WP0 lead			
Security officer			